

Part I: Midrise Housing

Midrise housing is housing that is at least three stories high. The design and construction of midrise housing varies significantly from that of low-rise housing. For example, steel frame wall and metal pan or solid concrete floor assemblies dominate midrise construction, while wood frame wall and wood underlayment floor assemblies dominate low-rise construction. One of the problems with midrise construction is that it is more difficult to diagnose air flow patterns and moisture migration in these large, multistory buildings.

The following two case studies describe a four-story development (constructed in 1990) located in Roxbury, Massachusetts, and a seven-story development (constructed in 1980) located in Holyoke, Massachusetts. The moisture problems in these structures included condensation caused by moist air transport against cold surfaces, as well as thermal bridging and rain intrusion through the building skin.

Family Housing, Roxbury Corners

The major complaint at Roxbury Corners Apartments was mold



At Roxbury Corners, we performed a rain leakage test to determine if rainwater was leaking into the walls.

growth. Mold was growing in every apartment, but primarily in the ground floor apartments and in the lower levels of the stairwells.

Roxbury Corners is four stories high, with steel frame construction. The exterior cladding is brick veneer over gypsum board. The

apartments range in size from two to four bedrooms and from 950 to 1,300 ft². A central gas-fired boiler located in the basement heats the apartments via hydronic baseboard fin tube radiators. Rooftop fans exhaust air from the kitchens and bathrooms. Test results from air depressurization indicated extremely leaky (5–6 ACH at 50 Pa) construction details for multifamily apartments of this type, age, and construction.

Investigators concluded that the mold growth occurred at Roxbury Corners because warm, humid air drawn into the building during the summer was condensing on cold surfaces, especially at metal studs. Basement level apartments are susceptible to this problem because they experience substantial heat loss to the ground at the below-grade portions of the perimeter walls and the bottom plates of the interior



After the rain leakage test is performed on the building exterior, field engineers removed sections of the interior wall to see if water had leaked into the walls. Because it had not, we knew that condensation was the cause of the mold growth.

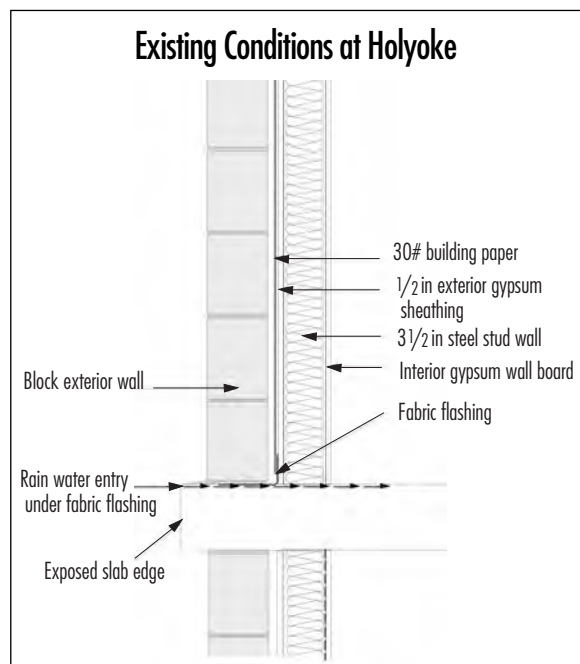


Figure 1. Rainwater entered the building, driven by wind, under the fabric flashing at the exterior balcony.



At Holyoke, the rain leakage test determined that rain water was indeed the moisture source.

walls. Warm, humid air enters into the hollow wall framing cavities and is cooled by the concrete foundation assembly. The mold growth occurs as a result of thermal bridging, which leads to a rise in the relative humidity of the air adjacent to the wall surface. The proposed solution to this problem was to increase the temperature of all cold surfaces, and to air seal at floor/exterior wall junctions and at interior unit perimeter walls to control air movement through the building.

Increase the surface temperature of cold surfaces. This measure required significant remediation and mitigation efforts in the lower-floor apartments. Investigators recommended reconstructing the lower apartments by adding rigid thermal insulation to the exterior walls in order to eliminate the thermal bridging. This required management to strip the damaged gypsum board and place rigid insulation between the metal framing and the concrete foundations at the perimeter.

The building management carried out these recommendations. All of the apartments were

inspected afterwards; the inspection included opening the walls and confirming that they are mold-free. The tenants at Roxbury Corners also no longer report mold growth in their homes.

Elderly Housing, Holyoke

The major cause of moisture problems at this development (constructed in 1980) was water

infiltration through the building skin. The complaints were about visible sources of water infiltration, mold, window condensation, window deterioration, and interior paint and gypsum wallboard damage.

Holyoke's midrise development

includes seven stories of small studio and one-bedroom flats that range in size from 600 to 800 ft². The building structure consists of a concrete frame with block infill walls and metal studs with gypsum board on both sides. A gas-fired central boiler heats water that circulates through the building to individual fan coil units. Exhaust fan systems in the kitchen and bathroom had been installed; however, they did not extract measurable quantities of air. Water leakage tests performed at the site documented water leakage through the facade. Air leakage tests documented fairly airtight construction (2 ACH at 50

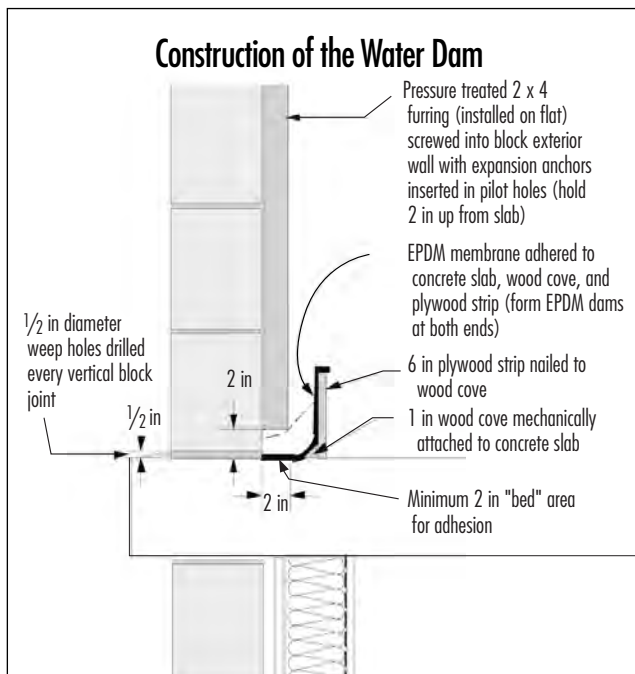


Figure 2. The existing stud wall and insulation were removed from the inside, and weep holes were drilled to allow the water to drain out.

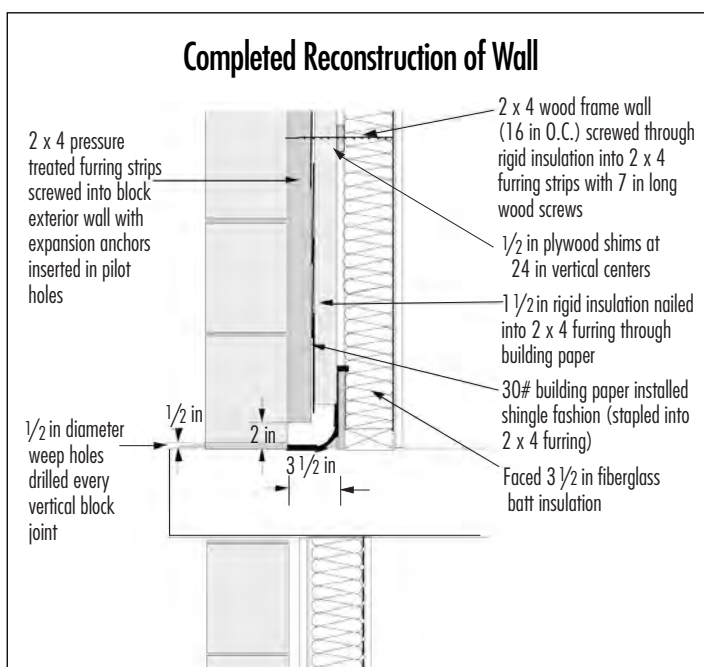


Figure 3. Once the water dam was installed, the interior wall was reconstructed. With the moisture source removed, the mold disappeared.

Pa, 1½ square inches leakage/100 ft² surface area).

Water leakage test results showed that rainwater was entering at the slab edge of each floor under the fabric flashing. At the same time, thermal bridging at the ceiling/exterior wall intersection was increasing the surface relative humidity in these locations high enough to support mold growth.



JOSEPH LISTBUREK

When the wall section at Holyoke was removed, water (and mold) were found within.

The recommended solution was to reduce the interior vapor pressure and reduce water infiltration through the exterior building envelope.

Reduce the interior vapor pressure. Install higher-capacity individual exhaust fans in each apartment.

Reduce building envelope water infiltration. Upgrade the building envelope. Figure 1 shows how water entered through the wall. Figure 2 shows a detail of the water dam and drainage path design recommended for locations where water entered the building. Figure 3 shows details of the solution to eliminate the thermal bridge effect of the exposed concrete frame. After these recommendations were implemented, water stopped entering the apartments.



*Part II of this article, Veteran's Era Housing, will appear in the next issue of **Home Energy**.*